Anand Ratna Maurya

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EDUCATION

Indian Institute of Technology, Madras *2021 – 2025*
Bachelor of Science in Data Science Chennai, India

EXPERIENCE

AI Model Trainer (Freelance) | Outlier AI *Jan 2024 – Apr 2025*

- Delivered 5+ end-to-end AI model training projects involving Supervised Fine-Tuning (SFT), model evaluation, and response ranking across Data Science, Machine Learning, Mathematics, Python, and SQL domains.
- Collaborated with 1,000+ global experts to develop structured training datasets, preference annotations, and evaluation workflows through extensive research, information gathering, and quality review processes.
- Performed prompt engineering, quality assurance, and error analysis on Google Compute Engine infrastructure, supporting model alignment, decision-making, knowledge retention, and continuous improvement through dedicated evaluation.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, Bash, Git, Docker
ML/DL: Scikit-learn, XGBoost, PyTorch, TensorFlow, Feature Engineering, Model Evaluation, Statistical Modeling
LLMs & Generative AI: RAG, SFT, Fine-Tuning, Hugging Face, Prompt Engineering, LangChain, LangGraph, smolagents
MLOps & Deployment: FastAPI, Docker, MLflow, Airflow, DVC, GitHub Actions, CI/CD, Model Serving
Cloud & Vector Databases: GCP, ChromaDB, FAISS

PROJECTS

BlackCipher AI – AI-Powered Cyber Threat Intelligence Platform
BlackCipher AI Repository

- Developed an end-to-end cyber threat intelligence platform combining Machine Learning, Explainable AI (SHAP), RAG, MITRE ATT&CK mapping, and LLM-based security analysis for automated threat detection and investigation.
- Developed an XGBoost-based intrusion detection model on the UNSW-NB15 dataset, achieving a 0.98 Weighted F1 Score for multiclass cyber threat detection.
- Built an intelligent RAG pipeline using FAISS, Sentence Transformers, and local Ollama LLMs (TinyLlama) to generate contextual threat intelligence summaries, SOC-style incident reports, risk assessments, and mitigation recommendations.
- Engineered a production-style solution using FastAPI, SHAP explainability, and MITRE ATT&CK mapping, enabling real-time threat analysis, feature attribution, attack classification, and API-driven cybersecurity workflows.

ChronosAI – Autonomous Root Cause Analysis Platform
ChronosAI Repository

- Developed an end-to-end Agentic AI platform for autonomous incident investigation, combining Multi-Agent Systems, RAG, LangGraph, and LLM-based reasoning workflows for root cause analysis, remediation planning, and operational decision-making.
- Architected a 6-agent autonomous workflow for incident classification, log analysis, metrics collection, knowledge retrieval, root cause analysis, and remediation planning.
- Built an intelligent RAG pipeline using FAISS, Sentence Transformers, and local Ollama LLMs, enabling contextual knowledge retrieval, structured investigations, and evidence-based analysis.
- Integrated 3 MCP servers, 7+ operational tools, and an 8-stage investigation pipeline using FastAPI, SQLite, and Docker for automated report generation, incident tracking, and operational planning.

Heart Disease MLOps Pipeline
Heartguard MLOps Repository

- Built an end-to-end MLOps pipeline for heart disease prediction using a Random Forest model with 13 clinical features, supporting automated training, evaluation, and production inference workflows.
- Developed a scalable FastAPI serving layer exposing real-time prediction APIs, containerized with Docker and deployed on GKE with Kubernetes HPA autoscaling (1-3 pods).
- Implemented ML observability through SHAP explainability, Fairlearn-based bias analysis, request logging, and drift detection that identified significant distribution shifts in cholesterol values.
- Automated deployment using GitHub Actions CI/CD and stress-tested the system under 200+ concurrent connections, achieving approximately 200+ requests/sec with 800 ms latency.

CORE COMPETENCIES

Multi-Agent AI Systems • Retrieval-Augmented Generation (RAG) • LLM Applications • LangGraph Workflows • MLOps & Model Serving • End-to-End ML Pipelines • Vector Search & Embeddings • FastAPI Development • Cloud Infrastructure (GCP) • CI/CD Automation • Feature Engineering • Applied Machine Learning